DAY – 09

ACTIVIY - 01

API TESTING USING POSTMAN

What is API?

APIs, or Application Programming Interfaces, are the invisible backbone of modern software development. They enable applications and systems to communicate and share data efficiently.

- Enable seamless interaction between applications, platforms, and services.
- Power everyday features such as weather updates, payments, and e-commerce checkouts.

What is Postman?

Postman is a tool that can be used for API Testing. It a widely used API testing tool that simplifies this process by providing a user-friendly interface for sending requests, validating responses, and automating test scripts.

Why <u>we are</u> using postman?

Postman is a widely used API development and testing platform that streamlines the process of designing, sending, and validating HTTP requests. It is essential for developers, testers, and DevOps teams to build, debug, document, and monitor APIs efficiently.

It supports all HTTP methods like GET, POST, PUT, PATCH, DELETE, and allows sending data in formats such as form-data, URL-encoded, raw JSON/XML, or binary. Postman also integrates with API authentication methods like API keys, OAuth, Bearer Tokens, and Basic Auth.

ACTIVITY – 02

DIFFERENCE BETWEEN WATERFALL , DEVOPS, AGILE

WATERFALL MODEL:

1. It is a linear and sequential software development model.
2. Each phase must be completed fully before starting the next phase.
3. No going back to previous phases once completed.
4. Phases follow a fixed order:
   a. Requirements
   b. Design
   c. Coding
   d. Testing
   e. Deployment
   f. Maintenance
5. Best suited for projects with clear and stable requirements.
6. Heavily documentation-based process.

AGILE

1. **Agile** is a way of working where project is split into many small sprints
2. That allows us to develop the software very faster in a better way with fewer mistakes.
3. We should get the feedback from the client after completing 1 or 2 sprints and this customer collaboration in between the projects considered as the big advantage in agile model, which helps the client to check the progress of the project often and to change the direction of the project easily when they needed.
4. Agile helps teams deliver work **faster, better, and with fewer mistakes**.

Phases of agile model:

1. Requirement Analysis
2. System Design
3. Implementation (Development)
4. Testing